CHAPTER 14
MATERIAL-SPECIFIC SEISMIC DESIGN AND DETAILING REQUIREMENTS

14.0 SCOPE

Structural elements including foundation elements shall conform to the material design and detailing requirements set forth in this chapter or as otherwise specified for nonbuilding structures in Tables 15.4-1 and 15.4-2.

14.1 STEEL

Structures, including foundations, constructed of steel to resist seismic loads shall be designed and detailed in accordance with this standard including the reference documents and additional requirements provided in this section.

14.1.1 Reference Documents. The design, construction, and quality of steel members that resist seismic forces shall conform to the applicable requirements, as amended herein, of the following: ANSI/AISC 341, ANSI/AISC 360, ANSI/AISI S100, ANSI/AISI S230, ANSI/AISI S310, ANSI/AISI S400, ASCE 8, ASCE 19, ANSI/SDI-C, ANSI/SDI-NC, ANSI/SDI QA/QC, ANSI/SDI-RD, ANSI/SJI-CJ, and ANSI/SJI-100.

14.1.2 Structural Steel

14.1.2.1 General. The design of structural steel for buildings and structures shall be in accordance with ANSI/AISC 360. Where required, the seismic design of structural steel structures shall be in accordance with the additional provisions of Section 14.1.2.2.

14.1.2.2 Seismic Requirements for Structural Steel Structures. The design of structural steel structures to resist seismic forces shall be in accordance with the provisions of Section 14.1.2.2.1 or 14.1.2.2.2, as applicable.

14.1.2.2.1 Seismic Design Categories B and C. Structural steel structures assigned to Seismic Design Category B or C shall be of any construction permitted by the applicable reference documents in Section 14.1.1. Where a response modification coefficient, R, in accordance with Table 12.2-1 is used for the design of structural steel structures assigned to Seismic Design Category B or C, the structures shall be designed and detailed in accordance with the requirements of AISC 341.

EXCEPTION: The response modification coefficient, R, designated for "steel systems not specifically detailed for seismic resistance, excluding cantilever column systems" in Table 12.2-1 shall be permitted for systems designed and detailed in accordance with AISC 360 and need not be designed and detailed in accordance with AISC 341.

14.1.2.2.2 Seismic Design Categories D through F. Structural steel structures assigned to Seismic Design Category D, E, or F shall be designed and detailed in accordance with AISC 341, except as permitted in Table 15.4-1.

14.1.3 Cold-Formed Steel

14.1.3.1 General. The design of cold-formed carbon or low-alloy steel structural members shall be in accordance with the requirements of AISI S100, and the design of cold-formed stainless steel structural members shall be in accordance with the requirements of ASCE 8. Where required, the seismic design of cold-formed steel structures shall be in accordance with the additional provisions of Section 14.1.3.2.

14.1.3.2 Seismic Requirements for Cold-Formed Steel Structures. Where a response modification coefficient, R, in accordance with Table 12.2-1, is used for the design of cold-formed steel structures, the structures shall be designed and detailed in accordance with the requirements of AISI S100, ASCE 8, and AISI S400, as applicable.

14.1.4 Cold-Formed Steel Light-Frame Construction

14.1.4.1 General. Cold-formed steel light-frame construction shall be designed in accordance with AISI S100, Section I4. Where required, the seismic design of cold-formed steel light-frame construction shall be in accordance with the additional provisions of Section 14.1.4.2.

14.1.4.2 Seismic Requirements for Cold-Formed Steel Light-Frame Construction. The design of cold-formed steel light-frame construction to resist seismic forces shall be in accordance with the provisions of Section 14.1.4.2.1 or 14.1.4.2.2, as applicable.

14.1.4.2.1 Seismic Design Categories B and C. Where a response modification coefficient, R, in accordance with Table 12.2-1 is used for the design of cold-formed steel light-frame construction assigned to Seismic Design Category B or C, the structures shall be designed and detailed in accordance with the requirements of AISI S400.

EXCEPTION: The response modification coefficient, R, designated for "steel systems not specifically detailed for seismic resistance, excluding cantilever column systems" in Table 12.2-1 shall be permitted for systems designed and detailed in accordance with AISI S100, Section I4 and need not be designed and detailed in accordance with AISI S400.

14.1.4.2.2 Seismic Design Categories D through F. Cold-formed steel light-frame construction structures assigned to Seismic Design Category D, E, or F shall be designed and detailed in accordance with AISI S400.

14.1.4.3 Prescriptive Cold-Formed Steel Light-Frame Construction. Cold-formed steel light-frame construction for one- and two-family dwellings is permitted to be designed and constructed in accordance with the requirements of AISI S230 subject to the limitations therein.

14.1.5 Cold-Formed Steel Deck Diaphragms. Cold-formed steel deck diaphragms shall be designed in accordance with the requirements of AISI S100, SDI-RD, SDI-NC, SDI-C or ASCE 8, as applicable. Nominal strengths shall be determined in accordance with AISI S310. The required strength of diaphragms, including bracing members that form part of the diaphragm, shall be determined in accordance with Section 12.10.1. Special inspections and qualification of welding special inspectors for cold-formed steel floor and roof deck shall be in accordance with the quality assurance inspection requirements of SDI-QA/QC.

14.1.6 Open Web Steel Joists and Joist Girders. The design, manufacture, and use of open web steel joists and joist girders shall be in accordance with SJI-100 and SJI-CJ, as applicable.

14.1.7 Steel Cables. The design strength of steel cables serving as main structural load carrying members shall be determined by the requirements of ASCE/SEI 19.

14.1.8 Additional Detailing Requirements for Steel Piles in Seismic Design Categories D through F. In addition to the foundation requirements set forth in Sections 12.1.5 and 12.13, design and detailing of H-piles shall conform to the requirements of AISC 341, and the connection between the pile cap and steel piles or unfilled steel pipe piles in structures assigned to Seismic Design Category D, E, or F shall be designed for a tensile force not less than 10% of the pile compression capacity.

EXCEPTION: Connection tensile capacity need not exceed the strength required to resist seismic load effects including overstrength of Section 12.4.3 or 12.14.3.2. Connections need not be provided where the foundation or supported structure does not rely on the tensile capacity of the piles for stability under the design seismic forces.

14.2 CONCRETE

Structures, including foundations, constructed of concrete to resist seismic loads shall be designed and detailed in accordance with this standard, including the reference documents and additional requirements provided in this section.

14.2.1 Reference Documents. The quality and testing of concrete materials and the design and construction of structural concrete members that resist seismic forces shall conform to the requirements of ACI 318, except as modified in Section 14.2.2.

14.2.2 Modifications to ACI 318. The text of ACI 318 shall be modified as indicated in Sections 14.2.2.1 through 14.2.2.7. Italics are used for text within Sections 14.2.2.1 through 14.2.2.7 to indicate requirements that differ from ACI 318.

14.2.2.1 Definitions. Add the following definitions to ACI 318, Section 2.3.

CONNECTION: *A region that joins two or more members. For precast concrete diaphragm design, a connection also refers to an assembly of connectors with the linking parts, welds and anchorage to concrete, which forms a load path across a joint between members, at least one of which is a precast concrete member.*

CONNECTOR: *Fabricated part embedded in concrete for anchorage and intended to provide a load path across a precast concrete joint.*

DETAILED PLAIN CONCRETE STRUCTURAL WALL: *A wall complying with the requirements of ACI, Chapter 14.*

ORDINARY PRECAST STRUCTURAL WALL: *A precast wall complying with the requirements of ACI 318 excluding Chapters 14, 18, and 27.*

PRECAST CONCRETE DIAPHRAGM DESIGN OPTIONS:

Basic Design Option (BDO): *An option where elastic diaphragm response in the design earthquake is targeted.*

Elastic Design Option (EDO): *An option where elastic diaphragm response in the maximum considered earthquake is targeted.*

Reduced Design Option (RDO): *An option that permits limited diaphragm yielding in the design earthquake.*

14.2.2.2 ACI 318, Section 10.7.6. Modify Section 10.7.6 by revising Section 10.7.6.1.6 to read as follows:

10.7.6.1.6 *If anchor bolts are placed in the top of a column or pedestal, the bolts shall be enclosed by transverse reinforcement that also surrounds at least four longitudinal bars within the column or pedestal. The transverse reinforcement shall be distributed within 5 in. of the top of the column or pedestal and shall consist of at least two No. 4 or three No. 3 bars. In structures assigned to Seismic Design Categories C, D, E, or F, the ties shall have a hook on each free end that complies with Section 25.3.4.*

14.2.2.3 Scope. Modify ACI 318, Section 18.2.1.2, to read as follows:

18.2.1.2 *All members shall satisfy requirements of Chapters 1 to 17 and 19 to 26. Structures assigned to SDC B, C, D, E, or F also shall satisfy Section 18.2.1.3 through 18.2.1.7, as applicable, except as modified by the requirements of Chapters 14 and 15 of ASCE 7. Where ACI 318, Chapter 18 conflicts with other ACI 318 chapters, Chapter 18 shall govern over those other chapters.*

14.2.2.4 Intermediate Precast Structural Walls. Modify ACI 318, Section 18.5, by renumbering Sections 18.5.2.2 and 18.5.2.3 to Sections 18.5.2.3 and 18.5.2.4, respectively, and adding new Section 18.5.2.2 to read as follows:

18.5.2.2 *Connections that are designed to yield shall be capable of maintaining 80% of their design strength at the deformation induced by design displacement, or shall use type 2 mechanical splices.*

18.5.2.3 *Elements of the connection that are not designed to yield shall develop at least 1.5* S_y.

18.5.2.4 *In structures assigned to SDC D, E, or F, wall piers shall be designed in accordance with Sections 18.10.8 or 18.14.*

14.2.2.5 Special Precast Structural Walls. Modify ACI 318, Section 18.11.2.1, to read as follows:

18.11.2.1 *Special structural walls constructed using precast concrete shall satisfy all requirements of Section 18.10 in addition to Section 18.5.2 as modified by Section 14.2.2 of ASCE 7.*

14.2.2.6 Foundations. Modify ACI 318, Section 18.13.1.1, to read as follows:

18.13.1.1 *This section, as modified by Sections 12.1.5, 12.13, or 14.2 of ASCE 7, shall apply to foundations resisting earthquake-induced forces between structure and ground in structures assigned to SDC D, E, or F.*

14.2.2.7 Detailed Plain Concrete Shear Walls. Modify ACI 318, Section 14.6, by adding a new Section 14.6.2 to read

14.6.2 Detailed Plain Concrete Shear Walls

14.6.2.1 *Detailed plain concrete shear walls are walls conforming to the requirements for ordinary plain concrete shear walls and Section 14.6.2.2.*

14.6.2.2 Reinforcement shall be provided as follows:

a. *Vertical reinforcement of at least 0.20 in.^{2} (129 mm^2) in cross-sectional area shall be provided continuously from support to support at each corner, at each side of each opening, and at the ends of walls. The continuous vertical bar required beside an opening is permitted to substitute for the No. 5 bar required by Section 14.6.1.*
b. *Horizontal reinforcement of at least 0.20 in.^{2} (129 mm^2) in cross-sectional area shall be provided:*
   1. *continuously at structurally connected roof and floor levels and at the top of walls;*
   2. *at the bottom of load-bearing walls or in the top of foundations where doweled to the wall; and*
   3. *at a maximum spacing of 120 in. (3,048 mm).*

Reinforcement at the top and bottom of openings, where used in determining the maximum spacing specified in Item 3 in the preceding text, shall be continuous in the wall.

14.2.3 Additional Detailing Requirements for Concrete Piles. In addition to the foundation requirements set forth in Sections 12.1.5 and 12.13 of this standard and in Section 14.2.3 of ACI 318, design, detailing, and construction of concrete piles shall conform to the requirements of this section.

14.2.3.1 Concrete Pile Requirements for Seismic Design Category C. Concrete piles in structures assigned to Seismic Design Category C shall comply with the requirements of this section.

14.2.3.1.1 Anchorage of Piles. All concrete piles and concrete-filled pipe piles shall be connected to the pile cap by embedding the pile reinforcement in the pile cap for a distance equal to the development length as specified in ACI 318 as modified by Section 14.2.2 of this standard or by the use of field-placed dowels anchored in the concrete pile. For deformed bars, the development length is the full development length for compression or tension, in the case of uplift, without reduction in length for excess area.

Hoops, spirals, and ties shall be terminated with seismic hooks as defined in Section 2.3 of ACI 318.

Where a minimum length for reinforcement or the extent of closely spaced confinement reinforcement is specified at the top of the pile, provisions shall be made so that those specified lengths or extents are maintained after pile cutoff.

14.2.3.1.2 Reinforcement for Uncased Concrete Piles (SDC C). Reinforcement shall be provided where required by analysis. For uncased cast-in-place drilled or augered concrete piles, a minimum of four longitudinal bars, with a minimum longitudinal reinforcement ratio of 0.0025 and transverse reinforcement, as defined below, shall be provided throughout the minimum reinforced length of the pile as defined below starting at the top of the pile. The longitudinal reinforcement shall extend beyond the minimum reinforced length of the pile by the tension development length. Transverse reinforcement shall consist of closed ties (or equivalent spirals) with a minimum 3/8-in. (9-mm) diameter. Spacing of transverse reinforcing shall not exceed 6 in. (150 mm) or 8 longitudinal-bar diameters within a distance of three times the pile diameter from the bottom of the pile cap. Spacing of transverse reinforcing shall not exceed 16 longitudinal-bar diameters throughout the remainder of the minimum reinforced length.

The minimum reinforced length of the pile shall be taken as the greater of

1. One-third of the pile length;
2. A distance of 10 ft (3 m);
3. Three times the pile diameter; or
4. The flexural length of the pile, which shall be taken as the length from the bottom of the pile cap to a point where the concrete section cracking moment multiplied by a resistance factor of 0.4 exceeds the required factored moment at that point.

14.2.3.1.3 Reinforcement for Metal-Cased Concrete Piles (SDC C). Reinforcement requirements are the same as for uncased concrete piles.

EXCEPTION: Spiral-welded metal casing of a thickness not less than No. 14 gauge can be considered as providing concrete confinement equivalent to the closed ties or equivalent spirals required in an uncased concrete pile, provided that the metal casing is adequately protected from possible deleterious action because of soil constituents, changing water levels, or other factors indicated by boring records of site conditions.

14.2.3.1.4 Reinforcement for Concrete-Filled Pipe Piles (SDC C). Minimum reinforcement 0.01 times the cross-sectional area of the pile concrete shall be provided in the top of the pile with a length equal to two times the required cap embedment anchorage into the pile cap but not less than the development length in tension of the reinforcement.

14.2.3.1.5 Reinforcement for Precast Nonprestressed Piles (SDC C). A minimum longitudinal steel reinforcement ratio of 0.01 shall be provided for precast nonprestressed concrete piles. The longitudinal reinforcing shall be confined with closed ties or equivalent spirals of a minimum 3/8-in. (10-mm) diameter. Transverse confinement reinforcing shall be provided at a maximum spacing of eight times the diameter of the smallest longitudinal bar, but not to exceed 6 in. (152 mm), within three pile diameters of the bottom of the pile cap. Spacing of transverse reinforcement shall not exceed 6 in. (152 mm) throughout the remainder of the pile.

14.2.3.1.6 Reinforcement for Precast Prestressed Piles (SDC C). For the upper 20 ft (6 m) of precast prestressed piles, the minimum volumetric ratio of spiral reinforcement shall not be less than 0.007 or the amount required by the following equation:

$$\rho_s = \frac{0.12 f'_c}{f_{yh}} \tag{14.2-1}$$

where

ρ_s = volumetric ratio (vol. spiral/vol. core);
f'_c = specified compressive strength of concrete, psi (MPa); and
f_{yh} = specified yield strength of spiral reinforcement, which shall not be taken as greater than 85,000 psi (586 MPa).

A minimum of one-half of the volumetric ratio of spiral reinforcement required by Eq. (14.2-1) shall be provided for the remaining length of the pile.

14.2.3.2 Concrete Pile Requirements for Seismic Design Categories D through F. Concrete piles in structures assigned to Seismic Design Category D, E, or F shall comply with Section 14.2.3.1.1 and the requirements of this section.

14.2.3.2.1 Site Class E or F Soil. Where concrete piles are used in Site Class E or F, they shall have transverse reinforcement in accordance with Sections 18.7.5.2 through 18.7.5.4 of ACI 318 within seven pile diameters of the pile cap and of the interfaces between strata that are hard or stiff and strata that are liquefiable or are composed of soft to medium stiff clay.

14.2.3.2.2 Nonapplicable ACI 318 Sections for Grade Beam and Piles. ACI 318, Section 18.13.3.3, need not apply to grade beams designed to resist the seismic load effects including overstrength of Section 12.4.3 or 12.14.3.2. ACI 318, Section 18.13.4.3(a), need not apply to concrete piles. ACI 318, Section 18.13.4.3(b), need not apply to precast, prestressed concrete piles.

14.2.3.2.3 Reinforcement for Uncased Concrete Piles (SDC D through F). Reinforcement shall be provided where required by analysis. For uncased cast-in-place drilled or augered concrete piles, a minimum of four longitudinal bars with a minimum longitudinal reinforcement ratio of 0.005 and transverse confinement reinforcement in accordance with ACI 318, Sections 18.7.5.2 through 18.7.5.4 shall be provided throughout the minimum reinforced length of the pile as defined below starting at the top of the pile. The longitudinal reinforcement shall extend beyond the minimum reinforced length of the pile by the tension development length.

The minimum reinforced length of the pile shall be taken as the greatest of

1. One-half of the pile length;
2. A distance of 10 ft (3 m);
3. Three times the pile diameter; or
4. The flexural length of the pile, which shall be taken as the length from the bottom of the pile cap to a point where the concrete section cracking moment multiplied by a resistance factor of 0.4 exceeds the required factored moment at that point.

In addition, for piles located in Site Classes E or F, longitudinal reinforcement and transverse confinement reinforcement, as described above, shall extend the full length of the pile.

Where transverse reinforcing is required, transverse reinforcing ties shall be a minimum of No. 3 bars for up to 20-in. (500-mm) diameter piles and No. 4 bars for piles of larger diameter.

In Site Classes A through D, longitudinal reinforcement and transverse confinement reinforcement, as defined above, shall also extend a minimum of seven times the pile diameter above and below the interfaces of soft to medium stiff clay or liquefiable strata except that transverse reinforcing not located within the minimum reinforced length shall be permitted to use a transverse spiral reinforcement ratio of not less than one-half of that required in ACI 318, Section 18.7.5.4(a). Spacing of transverse reinforcing not located within the minimum reinforced length is permitted to be increased but shall not exceed the least of the following:

1. 12 longitudinal bar diameters,
2. One-half the pile diameter, and
3. 12 in. (300 mm).

14.2.3.2.4 Reinforcement for Metal-Cased Concrete Piles (SDC D through F). Reinforcement requirements are the same as for uncased concrete piles.

EXCEPTION: Spiral-welded metal casing of a thickness not less than No. 14 gauge can be considered as providing concrete confinement equivalent to the closed ties or equivalent spirals required in an uncased concrete pile, provided that the metal casing is adequately protected against possible deleterious action because of soil constituents, changing water levels, or other factors indicated by boring records of site conditions.

14.2.3.2.5 Reinforcement for Precast Nonprestressed Piles (SDC D through F). Transverse confinement reinforcement consisting of closed ties or equivalent spirals shall be provided in accordance with ACI 318, Sections 18.7.5.2 through 18.7.5.4, for the full length of the pile.

EXCEPTION: In other than Site Classes E or F, the specified transverse confinement reinforcement shall be provided within three pile diameters below the bottom of the pile cap, but it is permitted to use a transverse reinforcing ratio of not less than one-half of that required in ACI 318, Section 18.7.5.4(a), throughout the remainder of the pile length.

A minimum of four longitudinal bars, with a minimum longitudinal reinforcement ratio of 0.005, shall be provided throughout the minimum reinforced length of the pile as defined below starting at the bottom of the pile cap. The minimum reinforced length of the pile shall be taken as the greatest of the following:

1. One-half of the pile length;
2. A distance of 10 ft (3 m);
3. Three times the least pile dimension; and
4. The distance to a point where the concrete section cracking moment multiplied by a resistance factor 0.4 exceeds the required factored moment at that point.

14.2.3.2.6 Reinforcement for Precast Prestressed Piles (SDC D through F). In addition to the requirements for Seismic Design Category C, the following requirements shall be met:

1. Requirements of ACI 318, Chapter 18, need not apply.
2. Where the total pile length in the soil is 35 ft (10,668 mm) or less, the ductile pile region shall be taken as the entire length of the pile. Where the pile length exceeds 35 ft (10,668 mm), the ductile pile region shall be taken as the greater of 35 ft (10,668 mm) or the distance from the underside of the pile cap to the point of zero curvature plus three times the least pile dimension.
3. In the ductile pile region, the center-to-center spacing of the spirals or hoop reinforcement shall not exceed one-fifth of the least pile dimension, six times the diameter of the longitudinal strand, or 8 in. (203 mm), whichever is smallest.
4. Spiral reinforcement shall be spliced by lapping one full turn, by welding, or by the use of a mechanical connector. Where spiral reinforcement is lap-spliced, the ends of the spiral shall terminate in a seismic hook in accordance with ACI 318, except that the bend shall be not less than 135 deg. Welded splices and mechanical connectors shall comply with ACI 318, Section 25.5.7.
5. Where the transverse reinforcement consists of spirals or circular hoops, the volumetric ratio of spiral transverse reinforcement in the ductile pile region shall comply with

$$\rho_s = 0.25\left(\frac{f'_c}{f_{yh}}\right)\left(\frac{A_g}{A_{ch}} - 1.0\right)\left[0.5 + \frac{1.4P}{f'_c A_g}\right]$$

but not less than

$$\rho_s = 0.12\left(\frac{f'_c}{f_{yh}}\right)\left[0.5 + \frac{1.4P}{f'_c A_g}\right]$$

and

ρ_s need not exceed 0.021

where

ρ_s = volumetric ratio (vol. of spiral/vol. of core);

$f_c' \leq 6000$ psi (≤ 41.4 MPa);

f_{yh} = yield strength of spiral reinforcement ≤ 85 ksi (≤ 586 MPa);

A_g = pile cross-sectional area [in.^{2} (mm^2)];

A_{ch} = core area defined by spiral outside diameter [in.^{2} (mm^2)]; and

P = axial load on pile resulting from the load combination $1.2D + 0.5L + 1.0E$, lb (kN).

This required amount of spiral reinforcement is permitted to be obtained by providing an inner and outer spiral.

6. Where transverse reinforcement consists of rectangular hoops and crossties, the total cross-sectional area of lateral transverse reinforcement in the ductile region with spacing, s, and perpendicular to dimension, h_c, shall conform to

$$A_{sh} = 0.3sh_c\left(\frac{f_c'}{f_{yh}}\right)\left(\frac{A_g}{A_{ch}} - 1.0\right)\left[0.5 + \frac{1.4P}{f_c'A_g}\right]$$

but not less than

$$A_{sh} = 0.12sh_c\left(\frac{f_c'}{f_{yh}}\right)\left[0.5 + \frac{1.4P}{f_c'A_g}\right]$$

where

s = spacing of transverse reinforcement measured along length of pile [in. (mm)];

h_c = cross-sectional dimension of pile core measured center to center of hoop reinforcement [in. (mm)]; and

$f_{yh} \leq 70$ ksi (≤ 483 MPa).

The hoops and crossties shall be equivalent to deformed bars not less than No. 3 in size. Rectangular hoop ends shall terminate at a corner with seismic hooks.

7. Outside of the ductile pile region, the spiral or hoop reinforcement with a volumetric ratio not less than one-half of that required for transverse confinement reinforcement shall be provided.

14.2.4 Additional Design and Detailing Requirements for Precast Concrete Diaphragms. In addition to the requirements for reinforced concrete set forth in this standard and ACI 318, Chapter 6 and Section 18.12, design, detailing, and construction of diaphragms constructed with precast concrete components in SDC C, D, E, and F, or in SDC B and using the requirements of Section 12.10.3, shall conform to the requirements of this section.

14.2.4.1 Diaphragm Seismic Demand Levels. A diaphragm seismic demand level for each structure shall be determined, based on Seismic Design Category; number of stories, N; diaphragm span, L, as defined in Section 14.2.4.1.1; and diaphragm aspect ratio, AR, as defined in Section 14.2.4.1.2. For structures assigned to SDC B or C, the seismic demand level shall be designated as low. For structures assigned to SDC D, E, or F, the seismic demand level shall be determined in accordance with Fig. 14.2-1 and the following:

1. If AR is greater than or equal to 2.5 and the diaphragm seismic demand is low according to Fig. 14.2-1, the diaphragm seismic demand level shall be changed from low to moderate.
2. If AR is less than 1.5 and the diaphragm seismic demand is high according to Fig. 14.2-1, the diaphragm seismic demand level shall be changed from high to moderate.

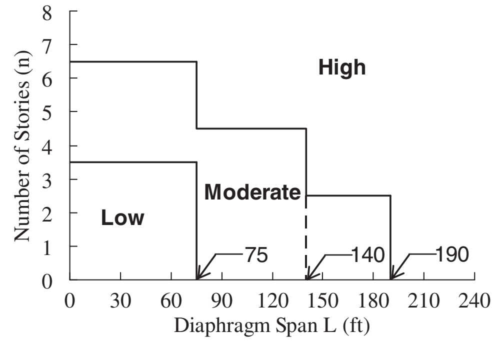

FIGURE 14.2-1 Diaphragm Seismic Demand Level

14.2.4.1.1 Diaphragm Span. Diaphragm span of a structure, L, shall be the maximum diaphragm span on any floor in the structure in any direction. The diaphragm span in a particular direction on a particular floor level shall be the larger of the maximum distance between two lateral force resisting system (LFRS) elements and twice the exterior distance between the outer LFRS element and the building free edge.

14.2.4.1.2 Diaphragm Aspect Ratio. The diaphragm aspect ratio, AR, shall be the diaphragm span-to-depth ratio using the diaphragm span, L, defined in Section 14.2.4.1.1. The diaphragm depth shall be the diaphragm dimension perpendicular to the diaphragm span between the chord lines for the diaphragm or portion of diaphragm.

14.2.4.1.3 Diaphragm Shear Amplification Factor. The required shear strength for diaphragm shall be amplified by the diaphragm shear overstrength factor, Ω_v, which shall be taken equal to $1.4R_s$.

14.2.4.2 Diaphragm Design Options. A diaphragm design option, as defined in Section 14.2.2.1, shall be assigned based on the lowest classification of connector or joint reinforcement deformability used.

14.2.4.2.1 Elastic Design Option. Any classification of connector or joint reinforcement deformability is permitted to be used with the elastic design option, which in turn is permitted for the following:

1. low seismic demand level and
2. moderate seismic demand level, provided that the diaphragm design force is increased 15%.

14.2.4.2.2 Basic Design Option. Either moderate deformability elements (MDEs) or high deformability elements (HDEs) shall be used with the basic design option, which is permitted for the following:

1. low seismic demand level,
2. moderate seismic demand level, and
3. high seismic demand level, provided that the diaphragm design force is increased 15%.

14.2.4.2.3 Reduced Design Option. High deformability elements (HDEs) shall be used with the reduced design option, which is permitted to be used for all seismic demand levels.

14.2.4.3 Diaphragm Connector or Joint Reinforcement Deformability. Precast concrete diaphragm connectors or joint reinforcement shall be classified in accordance with this section.

14.2.4.3.1 Low Deformability Element (LDE). Connectors or joint reinforcement used in precast concrete diaphragms with tension deformation capacity, as determined in Section 14.2.4.6.7, less than 0.3 in. (7.5 mm) are classified as low deformability elements.

14.2.4.3.2 Moderate Deformability Element (MDE). Connectors or joint reinforcement used in precast concrete diaphragms with tension deformation capacity, as determined in Section 14.2.4.6.7, greater than or equal to 0.3 in. (7.5 mm) but less than 0.6 in. (15 mm) are classified as moderate deformability elements.

14.2.4.3.3 High Deformability Element (HDE). Connectors or joint reinforcement used in precast concrete diaphragms with tension deformation capacity, as determined in Section 14.2.4.6.7, greater than or equal to 0.6 in. (15 mm) are classified as high deformability elements.

14.2.4.3.4 Connector/Joint Reinforcement Classification. Classification of precast concrete diaphragm reinforcement or connector elements shall be determined by testing of individual elements following the cyclic testing protocols defined in Section 14.2.4.4.

14.2.4.3.5 Deformed Bar Reinforcement. Deformed bar reinforcement (ASTM A615 or ASTM A706) placed in cast-in-place concrete topping or cast-in-place concrete pour strips and satisfying the cover, lap, and development requirements of ACI 318 shall be deemed to qualify as high deformability elements (HDEs).

14.2.4.3.6 Special Inspection. For a precast concrete joint reinforcement or connector classified as a high deformability element (HDE), installation of the embedded parts and completion of the reinforcement or connection in the field shall be subject to continuous special inspection performed by qualified inspectors under the supervision of a licensed design professional.

14.2.4.4 Precast Concrete Diaphragm Connector and Joint Reinforcement Qualification Procedure. Precast concrete diaphragm connectors and joint reinforcement shall be assigned to a deformability classification based on tests. The testing shall establish the strength, stiffness, and deformation capacity of the element. As a minimum, in-plane shear tests and in-plane tension tests shall be conducted. The following procedure is deemed to satisfy the test requirements.

14.2.4.4.1 Test Modules. A test module shall consist of two concrete elements connected by joint reinforcement or a connector or connectors. A separate full-scale test module and a minimum number of tests shall be used for each characteristic of interest. Modules shall be fabricated at full scale. Test modules shall include a minimum edge distance of 2 ft (0.6 m) from each connector centerline. Additional reinforcement shall be used to prevent premature failure of the test module. The additional reinforcement shall not be placed in a way that would alter the performance of the connector. The geometry, reinforcing details, fabrication procedures, and material properties of the connections and connected concrete elements shall be representative of those to be used in the prototype structure.

14.2.4.4.2 Number of Tests. Evaluation of test results shall be made on the basis of the values obtained from not fewer than three tests, provided that the deviation of any value obtained from any single test does not vary from the average value for all tests by more than 15%. If such deviation from the average value for any test exceeds 15%, then additional tests shall be performed until the deviation of any test from the average value does not exceed 15% or a minimum of six tests has been performed. No test shall be eliminated unless a rationale for its exclusion is given.

14.2.4.4.3 Test Configuration. For each connection test, a multidirectional test fixture shall be used to allow for the simultaneous control of shear, axial, and potential bending deformations at the test module joint. Demand shall be applied through displacement control of up to three actuators. The test module shall be connected to restraint beams along each edge parallel to the joint; slip between the test module and beams shall be minimized. One support beam shall be fastened to the laboratory floor, providing a fixed edge, while the other beam shall rest on a low-friction movable support. Vertical movement of the panel shall be restricted.

14.2.4.4.4 Instrumentation. At a minimum, instrumentation shall consist of displacement and force transducers. Force shall be measured in line with each actuator to quantify shear and axial demands on the connection. To accommodate displacement control of the actuators, feedback transducers shall be incorporated into each actuator. Connection deformation shall be measured directly on the test module. A minimum of two axial transducers shall be used to determine the average axial opening and closing at the connection. Shear deformation shall be determined from measurements taken at the location of the connection. Transducer supports shall be placed on the test module at adequate distances from the connection to minimize damage to the transducer supports during the test.

14.2.4.4.5 Loading Protocols. Connections shall be loaded in in-plane shear and tension in accordance with the following:

1. Monotonic and cyclic tests shall be conducted under displacement control, using rates less than 0.05 in./s (1.25 mm/s). Each module shall be tested until its strength decreases to 15% of the maximum load.
2. A monotonic test shall be performed to determine the reference deformation, as defined in Section 14.2.4.4.6, Item 2, of the connector or reinforcement, if a reference deformation is not available. The test module shall be loaded under a monotonically increasing displacement until its strength decreases to 15% of the maximum load.
3. In-plane cyclic shear tests, with a constant 0.1-in. (2.5-mm) axial opening, shall be conducted to determine stiffness, strength, and deformation under shear loading. The test module shall be subject to increasing shear displacement amplitudes. Three fully reversed cycles shall be applied at each displacement amplitude. Starting from zero displacement, there shall be four increments of displacement amplitude equal to one-quarter of the reference displacement. This step shall be followed by two increments, each equal to one-half the reference displacement. Then there shall be two more increments, each equal to the reference displacement. This step shall be followed by increments equal to twice the reference displacement, until the strength decreases to 15% of the maximum load.
4. In-plane cyclic tension/compression tests shall be conducted to determine stiffness, strength, and deformation. Starting from zero displacement, there shall be four increments of tension displacement amplitudes equal to one-quarter of the reference displacement. This step shall be followed by two increments, each equal to one-half the reference displacement. Then there shall be two more increments, each equal to the reference displacement. This

step shall be followed by increments equal to twice the reference displacement, until the tensile strength decreases to 15% of the maximum load. There shall be three cycles of loading at each displacement amplitude. The compression portion of each cycle shall be force-limited. Each compression half cycle shall consist of an increasing compressive deformation until a force limit is reached. The force limit for each cycle shall be equal to the maximum force of the preceding tension half cycle. The shear deformation along the joint shall not be restrained during a tension/compression test.

14.2.4.4.6 Measurement Indices, Test Observations, and Acquisition of Data. The applied shear and tension/compression deformations and all resulting forces shall be recorded at least once every second and shall form the basis of Items 1 and 2:

1. Reference Deformation. The reference deformation, Δ_1, corresponding to Point 1, determined in Item 2, represents the effective yield deformation of the connector or reinforcement. An analytical determination of the reference deformation is permitted as an alternative to determination based on monotonic testing.
2. Backbone Qualification Envelope. The measured cyclic response shall be processed in accordance with the procedure below.

An envelope of the cyclic force deformation response shall be constructed from the force corresponding to the peak displacement applied during the first cycle of each increment of deformation. The envelope shall be simplified to a backbone curve consisting of four segments in accordance with Fig. 14.2-2.

Point 2 represents the peak envelope load. Point a is the point on the backbone curve where the strength first equals 15% of peak load. Initial elastic stiffness, K_e, shall be calculated as the slope of the secant of the strength-displacement relationship from origin to Point a. Point b is the point on the envelope curve at a displacement Δ_b. The displacement Δ_b is at the intersection of a horizontal line from the peak envelope load and the initial elastic stiffness line through 15% of the peak load. Point 1 represents the occurrence of yield, which is determined by drawing a line from Point 2 to b and extending it to intersect the initial elastic stiffness line through 15% of the peak load. Point 3 is the point where the strength has decreased to 15% of the peak load. Point 2a is the point where the deformation is 50% of the summation of deformations at Points 2 and 3.

The backbone curve shall be classified as one of the types indicated in Fig. 14.2-3. Deformation-controlled elements shall conform to Type 1 or Type 2, but not Type 2 Alternate, response with $\Delta_2 \geq 2\Delta_1$. All other responses shall be classified as force-controlled.

14.2.4.4.7 Response Properties. The following performance characteristics of the connector or joint reinforcement shall be quantified from the backbone response: the effective yield (reference) deformation, the tension deformation capacity, the tensile strength, and the shear strength—all determined as the average of values obtained from the number of tests required by Section 14.2.4.4.2. The tension deformation capacity shall correspond to Point 2, for deformation-controlled connections (see definition in Section 14.2.4.4.6, Item 2). It shall correspond to Point 1 for force-controlled connections, except that for force-controlled connections exhibiting Type 2 Alternate response, tension deformation capacity shall correspond to Point 1′.

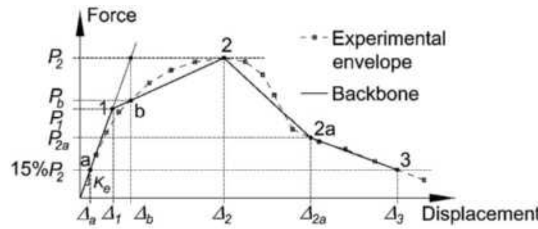

FIGURE 14.2-2 Backbone Qualification Curve

Deformation Category. The connector or joint reinforcement shall be classified as a low deformability element (LDE), a moderate deformability element (MDE), or a high deformability element (HDE) based on its deformation capacity in tension. The tension deformation capacity ranges given in Section 14.2.4.3 shall be used to determine the deformation category of the connector or reinforcement.

Tensile Strength. The tensile strength of the connector or joint reinforcement shall be the force corresponding to Point 1.

Shear Strength. If the shear deformation Δ_1 is less than 0.25 in. (6.4 mm), the shear strength shall be the force at Point 1. If the shear deformation Δ_1 is greater than or equal to 0.25 in. (6.4 mm), the shear strength shall be the force at 0.25 in. (6.4 mm) of shear deformation. This shear strength shall equal the stiffness, K_e, multiplied by 0.25 in. (6.4 mm).

14.2.4.4.8 Test Report. The test report shall be complete and self-contained for a qualified expert to be satisfied that the tests have been designed and carried out in accordance with the criteria previously described. The test report shall contain information enabling an independent evaluation of the performance of the test module. As a minimum, all of the following information shall be provided:

1. Details of test module design and construction, including engineering drawings.
2. Specified material properties used for design, and actual material properties obtained by testing.
3. Description of test setup, including diagrams and photographs.
4. Description of instrumentation, location, and purpose.
5. Description and graphical presentation of applied loading protocol.
6. Material properties of the concrete measured in accordance with ASTM C39. The average of a minimum of three tests shall be used. The compression tests shall be conducted within seven days of the connection tests or shall be interpolated from compression tests conducted before and after the connection test series.
7. Material properties of the connector, slug, and weld metal based on material testing or mill certification. As a minimum, the yield stress, tensile stress, and the ultimate strain shall be reported.
8. Description of observed performance, including photographic documentation, of test module condition at key deformation cycles.
9. Graphical presentation of force versus deformation response.
10. The envelope and backbone of the load-deformation response.
11. Yield strength, peak strength, yield deformation, tension deformation capacity, and connection deformation category.
12. Test date report date name of testing agency, report author(s), supervising professional engineer, and test sponsor.

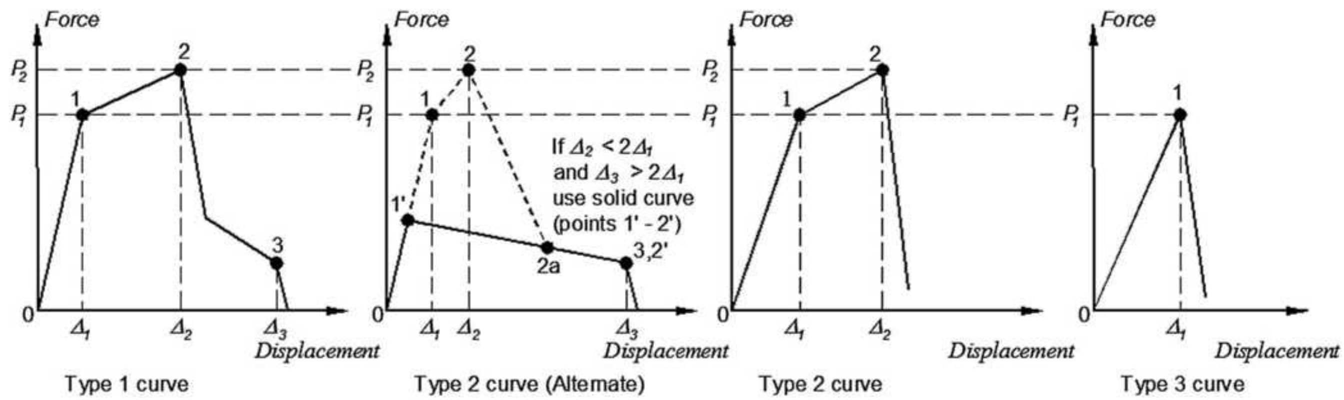

FIGURE 14.2-3 Deformation Curve Types

14.3 COMPOSITE STEEL AND CONCRETE STRUCTURES

Structures, including foundations, constructed of composite steel and concrete to resist seismic loads shall be designed and detailed in accordance with this standard, including the reference documents and additional requirements provided in this section.

14.3.1 Reference Documents. The design, construction, and quality of composite steel and concrete members that resist seismic forces shall conform to the applicable requirements of the following: AISC 341, AISC 360, and ACI 318, excluding Chapter 14.

14.3.2 General. Systems of structural steel acting compositely with reinforced concrete shall be designed in accordance with AISC 360 and ACI 318, excluding Chapter 14. Where required, the seismic design of composite steel and concrete systems shall be in accordance with the additional provisions of Section 14.3.3.

14.3.3 Seismic Requirements for Composite Steel and Concrete Structures. Where a response modification coefficient, *R*, in accordance with Table 12.2-1 is used for the design of systems of structural steel acting compositely with reinforced concrete, the structures shall be designed and detailed in accordance with the requirements of AISC 341.

14.3.4 Metal-Cased Concrete Piles. Metal-cased concrete piles shall be designed and detailed in accordance with Section 14.2.3.1.3.

14.4 MASONRY

Structures, including foundations, constructed of masonry to resist seismic loads shall be designed and detailed in accordance with this standard, including the references and additional requirements provided in this section.

14.4.1 Reference Documents. The design, construction, and quality assurance of masonry members that resist seismic forces shall conform to the requirements of TMS 402 and TMS 602, except as modified by Section 14.4.

14.4.2 *R* Factors. To qualify for the response modification coefficients, *R*, set forth in this standard, the requirements of TMS 402 and TMS 602, as amended in subsequent sections, shall be satisfied.

Special reinforced masonry shear walls designed in accordance with Section 8.3 or 9.3 of TMS 402 shall also comply with the additional requirements contained in Section 14.4.4 or 14.4.5.

14.4.3 Modifications to Chapter 7 of TMS 402

14.4.3.1 Separation Joints. Add the following new Section 7.5 to TMS 402:

__7.5.1 Separation Joints__. Where concrete abuts structural masonry and the joint between the materials is not designed as a separation joint, the concrete shall be roughened so that the average height of aggregate exposure is 1/8 in. (3 mm) and shall be bonded to the masonry in accordance with these requirements as if it were masonry. Vertical joints not intended to act as separation joints shall be crossed by horizontal reinforcement as required by Section 5.1.1.2.

14.4.4 Modifications to Chapter 6 of TMS 402

14.4.4.1 Reinforcement Requirements and Details

14.4.4.1.1 Reinforcing Bar Size Limitations. Modify the following within TMS 402, Section 6.1.2:

Delete TMS 402, Section 6.1.2.1, and replace with:

6.1.2.1 Reinforcing bars used in masonry shall not be larger than No. 9 (M#29).

Delete TMS 402, Section 6.1.2.2, and replace with:

6.1.2.2 The nominal bar diameter shall not exceed one-eighth of the nominal member thickness and shall not exceed one-quarter of the least clear dimension of the cell, course, or collar joint in which it is placed.

Add the following sentence to the end of TMS 402, Section 6.1.2.4:

The area of reinforcing bars placed in a cell or in a course of hollow unit construction shall not exceed 4% of the cell area.

14.4.4.1.2 Splices in Reinforcement. Add the following new Sections 6.1.6.1.1.4 and 6.1.6.1.2.1 to TMS 402:

6.1.6.1.1.4 Where $M/V_u d_v$ *exceeds 1.5 and the seismic load associated with the development of the nominal shear capacity exceeds 80% of the seismic load associated with development of the nominal flexural capacity, lap splices shall not be used in plastic hinge zones of special reinforced masonry shear walls. The length of the plastic hinge zone shall be taken as at least 0.15 times the distance between the point of zero moment and the point of maximum moment.*

6.1.6.1.2.1 Where $M/V_u d_v$ *exceeds 1.5 and the seismic load associated with the development of the nominal shear capacity exceeds 80% of the seismic load associated with* *development of the nominal flexural capacity, welded splices shall not be permitted in plastic hinge zones of special reinforced walls of masonry.*

Replace TMS 402, Section 6.1.6.1.3, as follows:

6.1.6.1.3 Mechanical Connections: Mechanical splices shall be classified as Type 1 or Type 2 according to Section 18.2.7.1 of ACI 318. Type 1 mechanical splices shall not be used within a plastic hinge zone or within a spandrel-pier joint of a special reinforced masonry shear wall system. Type 2 mechanical splices shall be permitted in any location within a member.

14.4.5 Modifications to Chapter 9 of TMS 402

14.4.5.1 Anchoring to Masonry. Add the following as the first paragraph in TMS 402, Section 9.1.6:

9.1.6 Anchor Bolts Embedded in Grout*. Anchorage assemblies connecting masonry elements that are part of the seismic force-resisting system to diaphragms and chords shall be designed so that the strength of the anchor is governed by steel tensile or shear yielding. Alternatively, the anchorage assembly is permitted to be designed so that it is governed by masonry breakout or anchor pullout provided that the anchorage assembly is designed to resist not less than 2.0 times the factored forces transmitted by the assembly.*

14.4.5.2 Coupling Beams. Add the following new Section 9.3.4.2.5 to TMS 402:

9.3.4.2.5 Coupling Beams*. Structural members that provide coupling between shear walls shall be designed to reach their moment or shear nominal strength before either shear wall reaches its moment or shear nominal strength. Analysis of coupled shear walls shall comply with accepted principles of mechanics.*

The design shear strength, ϕV_n, of the coupling beams shall satisfy the following criterion:

$$\phi V_n \geq \frac{1.25(M_1 + M_2)}{L_c} + 1.4V_g$$

where

M_1 *and* M_2 = *nominal moment strength at the ends of the beam;*
L_c = *length of the beam between the shear walls; and*
V_g = *unfactored shear force caused by gravity loads.*

The calculation of the nominal flexural moment shall include the reinforcement in reinforced concrete roof and floor systems. The width of the reinforced concrete used for calculations of reinforcement shall be six times the floor or roof slab thickness.

14.4.5.3 Walls with Factored Axial Stress Greater Than $0.05f'_m$. Add the following exception following the third paragraph of TMS 402, Section 9.3.5.4.2:

EXCEPTION: *A nominal thickness of 4 in. (102 mm) is permitted where load-bearing reinforced hollow clay unit masonry walls satisfy all of the following conditions:*

1. *The maximum unsupported height-to-thickness or length-to-thickness ratios do not exceed 27.*
2. *The net area unit strength exceeds 8,000 psi (55 MPa).*
3. *Units are laid in running bond.*
4. *Bar sizes do not exceed No. 4 (13 mm).*
5. *There are no more than two bars or one splice in a cell.*
6. *Joints are not raked.*

14.4.5.4 Shear Keys. Add the following new Section 9.3.6.7 to TMS 402:

9.3.6.7 Shear Keys. *The surface of concrete upon which a special reinforced masonry shear wall is constructed shall have a minimum surface roughness of 1/8 in. (3 mm). Shear keys are required where the calculated tensile strain in vertical reinforcement from in-plane loads exceeds the yield strain under load combinations that include seismic forces based on an* R *factor equal to 1.5. Shear keys that satisfy the following requirements shall be placed at the interface between the wall and the foundation:*

1. *The width of the keys shall be at least equal to the width of the grout space.*
2. *The depth of the keys shall be at least 1.5 in. (38 mm).*
3. *The length of the key shall be at least 6 in. (152 mm).*
4. *The spacing between keys shall be at least equal to the length of the key.*
5. *The cumulative length of all keys at each end of the shear wall shall be at least 10% of the length of the shear wall (20% total).*
6. *At least 6 in. (150 mm) of a shear key shall be placed within 16 in. (406 mm) of each end of the wall.*
7. *Each key and the grout space above each key in the first course of masonry shall be grouted solid.*

14.4.6 Modifications to Chapter 12 of TMS 402

14.4.6.1 Corrugated Sheet Metal Anchors. Add Section 12.2.2.11.1.1 to TMS 402 as follows:

12.2.2.11.1.1 *Provide continuous single wire joint reinforcement of wire size W1.7 (MW11) at a maximum spacing of 18 in. (457 mm) on center vertically. Mechanically attach anchors to the joint reinforcement with clips or hooks. Corrugated sheet metal anchors shall not be used.*

14.4.7 Modifications to TMS 602.

14.4.7.1 Construction Procedures. Add the following new Article 3.5 I to TMS 602:

3.5 I. *Construction procedures or admixtures shall be - used to facilitate placement and control shrinkage of grout.*

14.5 WOOD

Structures, including foundations, constructed of wood to resist seismic loads shall be designed and detailed in accordance with this standard including the references and additional requirements provided in this section.

14.5.1 Reference Documents. The quality, testing, design, and construction of members and their fastenings in wood systems that resist seismic forces shall conform to the requirements of the applicable following reference documents: AWC NDS and AWC SDPWS.

14.6 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

See Chapter 23 for the list of consensus standards and other documents that shall be considered part of this standard to the extent referenced in this chapter.

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